END-SEMESTER PROJECT · NOVEMBER 2026

ApexHunter

Finding driver mistakes and pit-stop windows in Formula 1 telemetry, and an honest test of how well it works

Parin Shah
23001091

92	57.9 M	2
sessions	readings	seasons

THE PROBLEM

Two questions asked of the car's data

1 Where did the driver make a mistake?

Moments where he did something different from the fastest drivers at the same spot, in a way that cost time: a lock-up, an early lift, a missed downshift.

2 When should he change tyres?

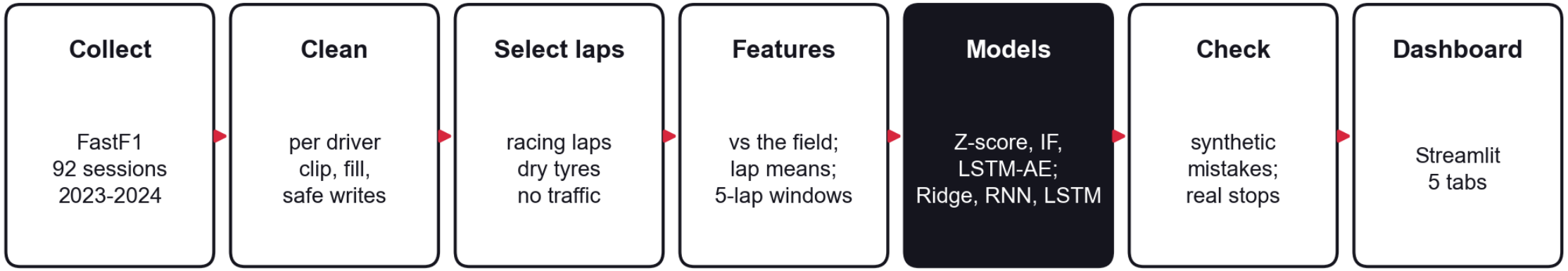
From the laps driven so far, a five-lap pit window, and a "pit now" call when a safety car makes the stop cheaper. Checked against the stops drivers really made.

Why it is hard:

No one labels mistakes, and a pit lap is a team decision, not a correct answer. The tyre signal is tiny: about 0.05 s per lap, against 0.9 s of normal lap-to-lap variation. So every model is tested on data it has not seen and compared with a simple baseline.

WHAT WE BUILT

From raw telemetry to a dashboard



Simple baseline

Z-score rule · "no change" (next lap = last lap)

Machine learning

Isolation Forest · Ridge regression

Deep learning

LSTM autoencoder · Vanilla RNN · LSTM

Data: every race and qualifying of 2023–2024 from FastF1: 92 sessions, 57.9 M readings, ~50,000 laps. 55 data and code problems were fixed first, each with a test.

QUESTION 1 · MISTAKES

Compare with the field, using three detectors

Key idea: braking before a corner is normal, braking on a straight is not. So each moment is compared with the 5 fastest other drivers at the same spot.

Five features: how much slower, and how different the throttle, brake, gear and acceleration are, compared with the field there.

Result: "mistakes" that were just braking fell from 95 % to about 45 %.

Z-score (baseline)

Flags a reading when any one feature is extreme.

Isolation Forest

Random trees isolate unusual readings in few splits.

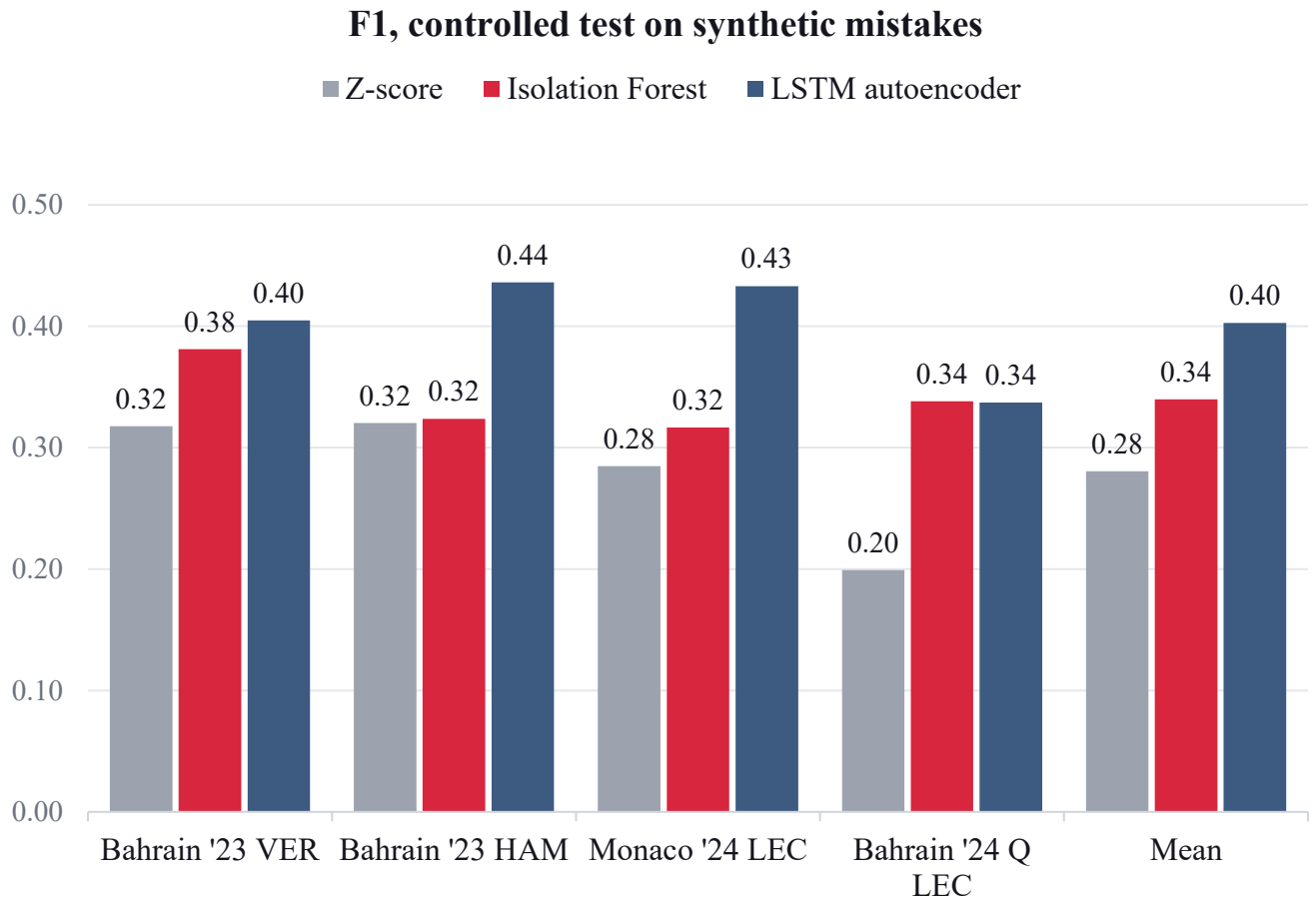
LSTM autoencoder

Learns to rebuild ~2 s of normal driving; what it cannot rebuild is unusual.

Two checks: a controlled test on synthetic mistakes (limit: shows the patterns are found, not that every real flag is a mistake) and a real-lap check (did a flag really cost time?)

QUESTION 1 · RESULTS

Both learned detectors beat the baseline



0.34

Isolation Forest mean F1 (Z-score 0.28)

0.40

LSTM autoencoder mean F1: finds the most synthetic mistakes

18 %

of Isolation Forest flags cost time on real laps (random places 12 %, autoencoder 14 %)

QUESTION 2 · TYRES

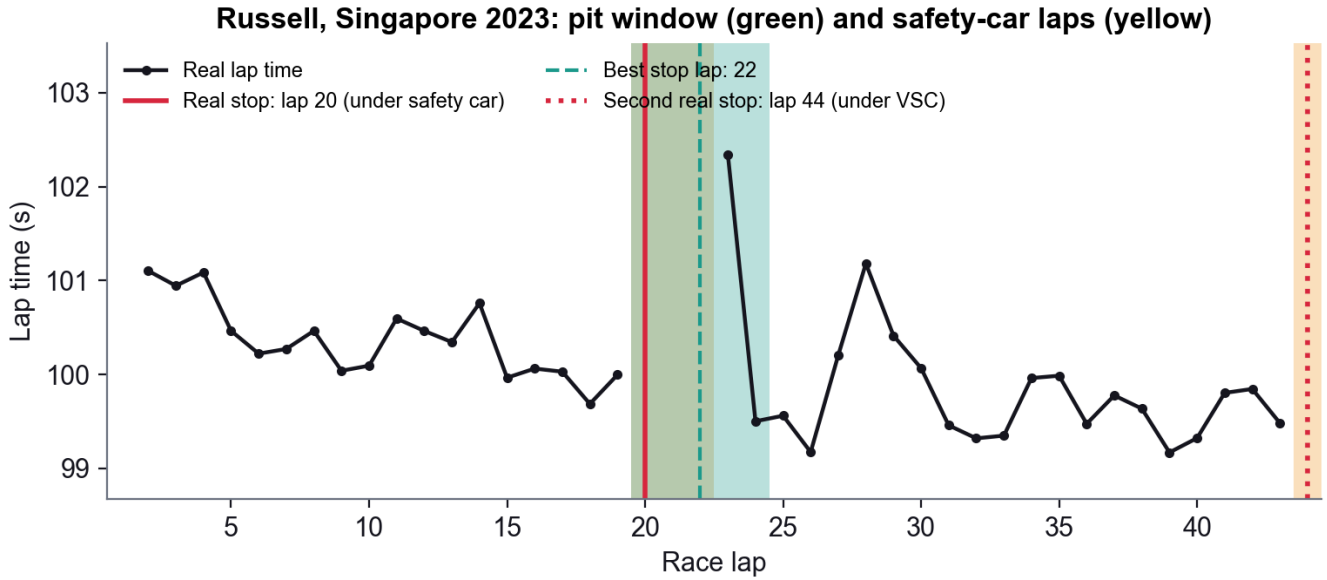
Predict the next lap, then find the pit window

The model sees the driver's last 5 laps: pace, throttle, brake, tyre age and compound.
It predicts how many seconds slower the next lap will be (the change, because a lap time mostly tells you the track).

A pit stop costs 22.8 s (median), 30 % less under a safety car.

The window in 3 steps:

1. estimate tyre wear per lap
2. for each possible stop lap, add tyre time lost + stop cost
3. cheapest lap ± 2 laps = the window



Example: Russell, Singapore 2023. Window laps 20–24. A safety car on lap 20 made the stop 30 % cheaper, so "pit now". He really pitted on lap 20, inside the window.

An LSTM autoencoder learns normal driving

1 Input

16 readings (~2 s) of the five deviation features

2 Encoder → decoder

an LSTM squeezes the window into 16 numbers, a second LSTM rebuilds it

3 Score

rebuild error; the 2 % worst readings are flagged, as for the Isolation Forest

0.40

mean F1 on synthetic mistakes, the best of the three (Isolation Forest 0.34)

0.34

recall: it finds most mistakes that unfold over time, such as early lifts

14 %

of its real flags cost time: about as often as random places (12 %)

Two recurrent networks read the five laps in order

1 Vanilla RNN

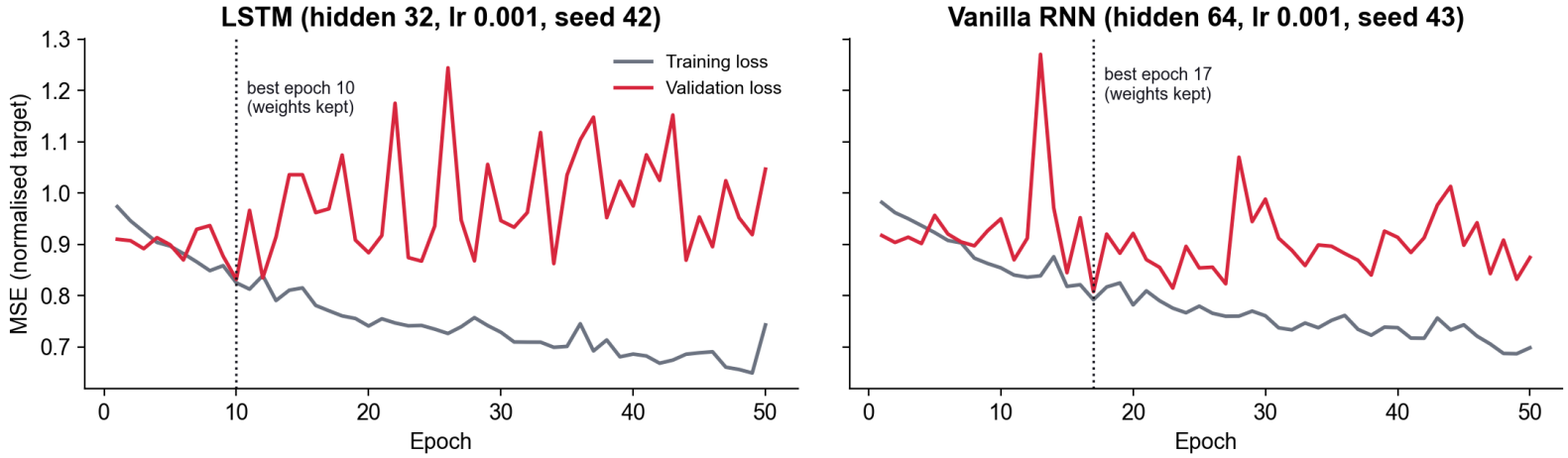
Reads one lap at a time and keeps a running memory; over long sequences that memory fades (vanishing gradients).

2 LSTM

Adds gates that decide what to remember, forget and pass on, so the memory does not fade.

3 Fair comparison

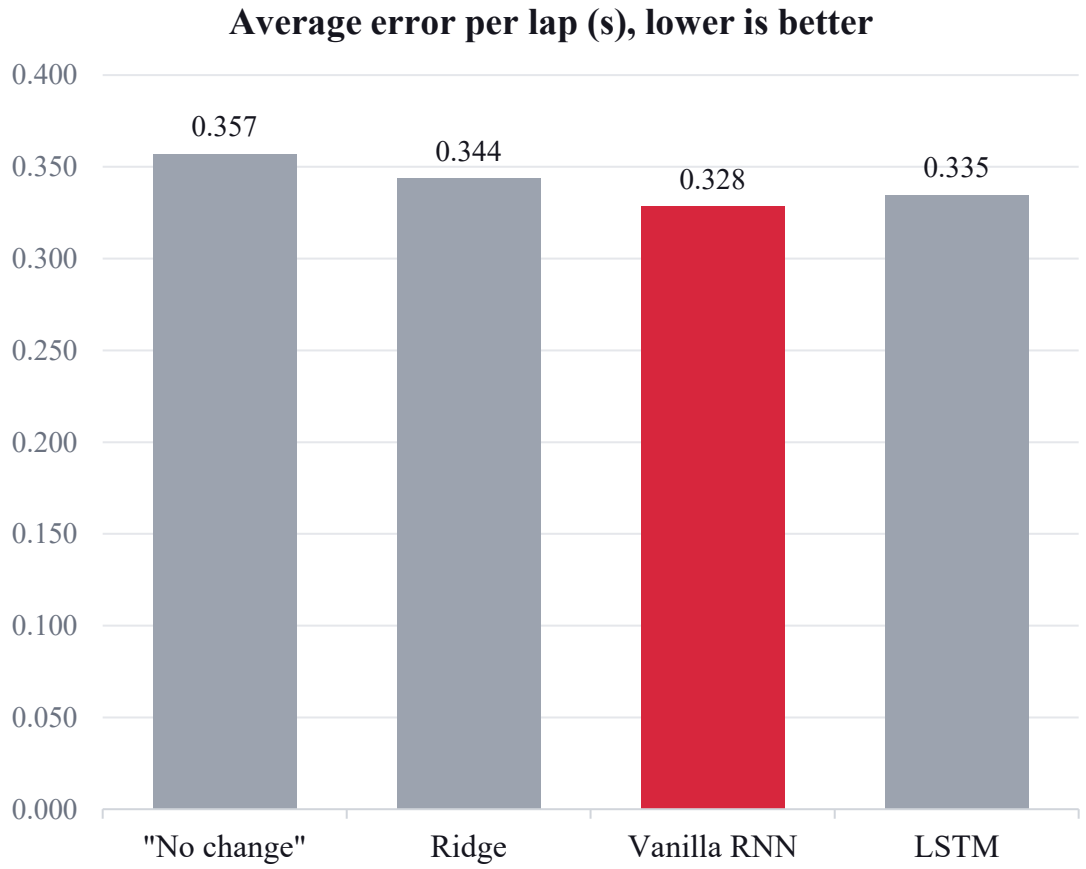
Same laps, same test races, same baseline; trained 3 times each and averaged.

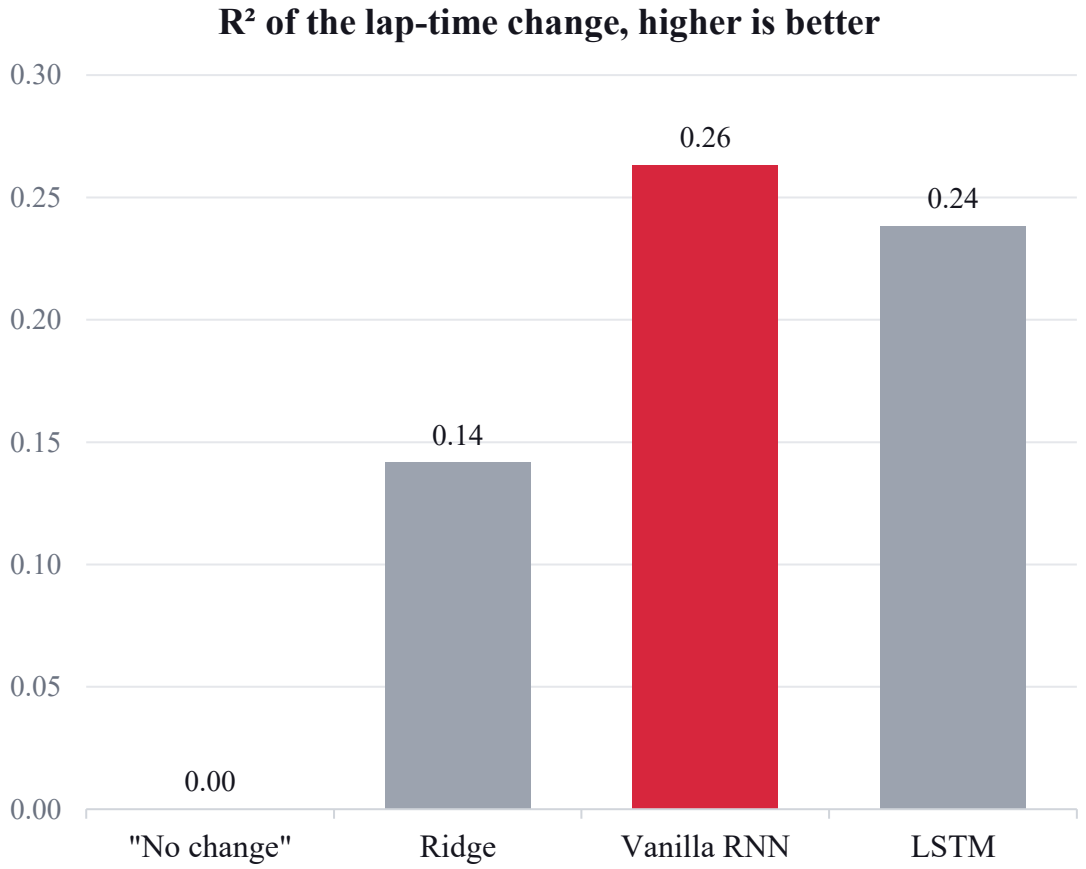


Overfitting, and the fix:
after 10–17 epochs validation error stops improving while training error keeps falling: the networks memorise noise. We keep the best validation epoch.

RESULTS · NEXT-LAP PREDICTION

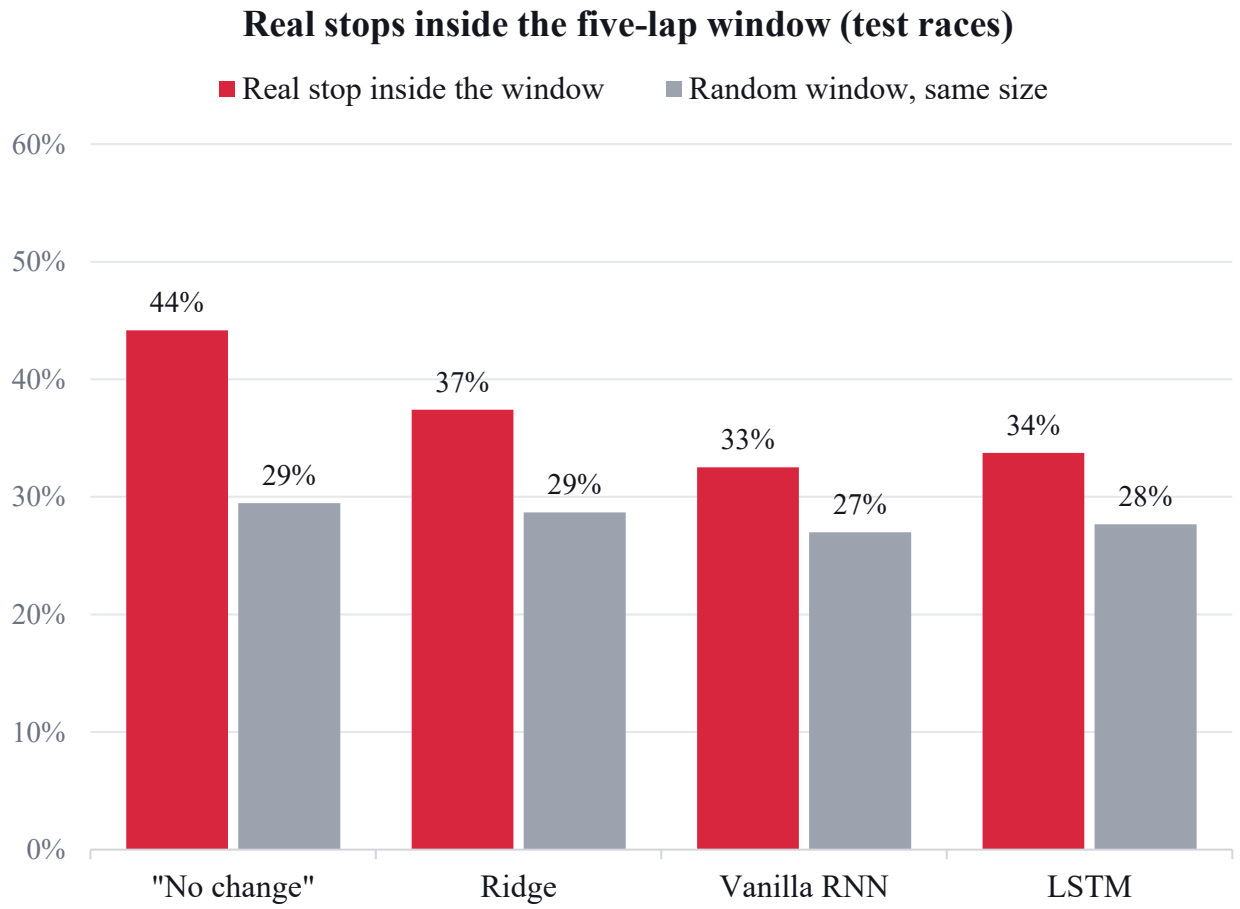
All models beat the baseline; the networks do best

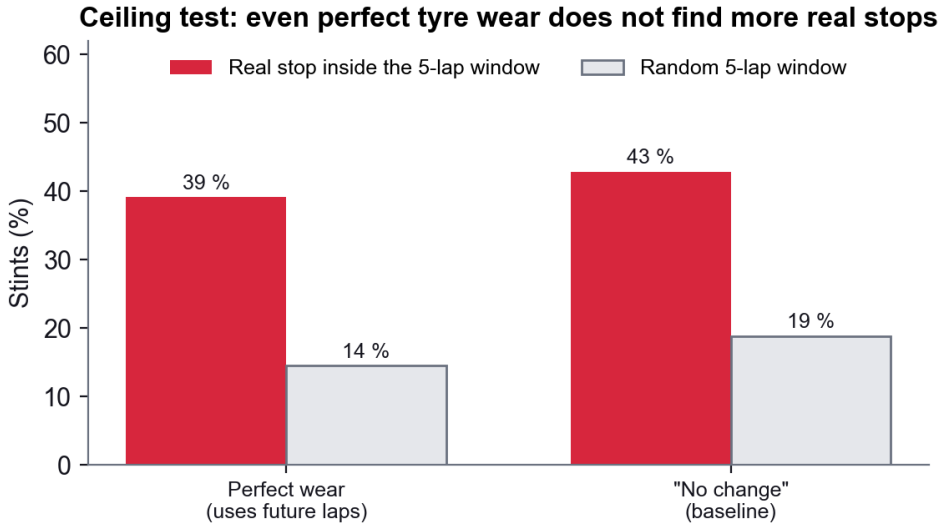




5 test races never seen in training. The RNN ties the LSTM: with only 5 laps, the LSTM's longer memory does not help.

Better lap times do not mean better pit windows



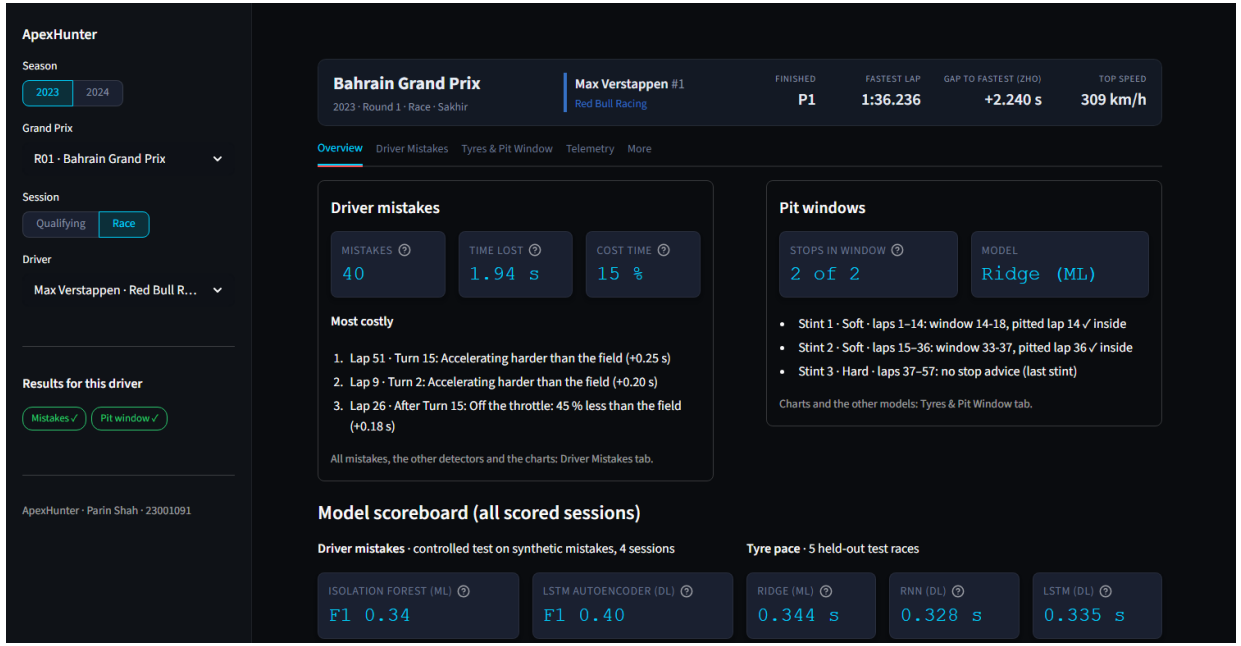


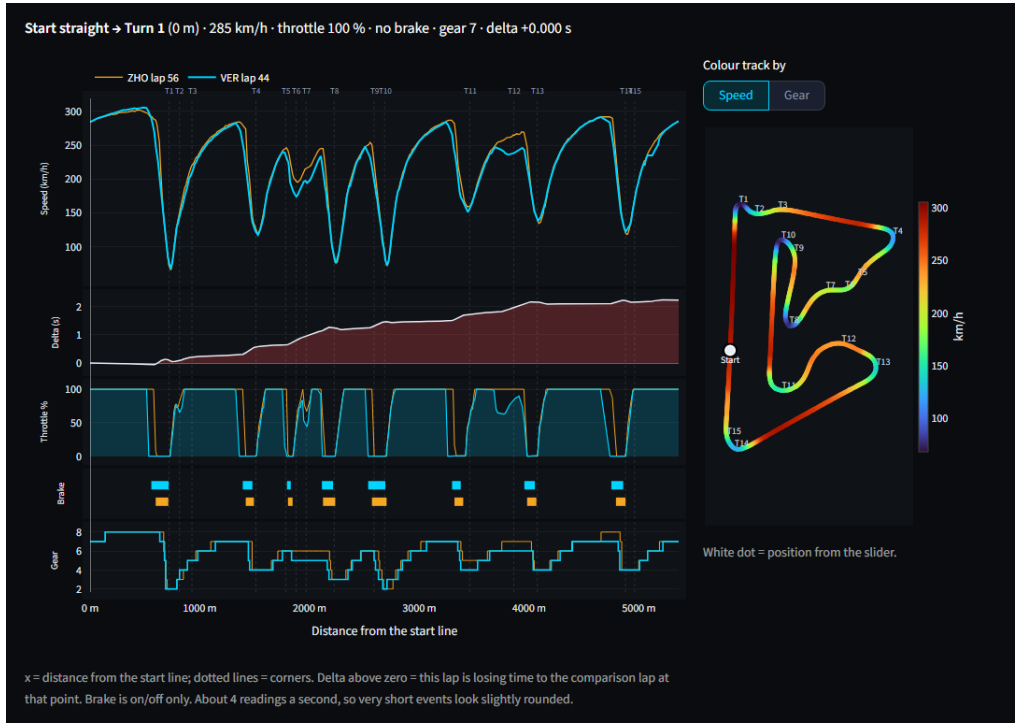
Every model beats a random window, but none beats the simple baseline.

Ceiling test: even perfect tyre wear (using future laps) finds no more real stops. Stops are decided by strategy, traffic and safety cars.

THE DASHBOARD

Five tabs, one chart per question





Overview

both answers + scoreboard

Driver Mistakes

3 detectors, map, list

Tyres & Pit Window

Ridge / RNN / LSTM

Telemetry

any lap, against distance

More

last semester's parts

WRAPPING UP

What we learned, and what we can't claim

What we learned

Clean the data first: 55 problems would have changed the results.

Compare with the right reference: the field at the same spot.

Test with an answer key, and check it on real laps.

Deep learning is not automatic: the plain RNN ties the LSTM.

Report what does not work: the ceiling test.

Limits and next steps

Synthetic mistakes show the patterns are found, not that every flag is a real mistake.

Only four sessions scored for mistakes; five test races for tyres.

The strategy ignores traffic and rivals on purpose.

Next: predict the laps until a driver pits; add traffic.

APEXHUNTER

Thank you

Questions are welcome

Mistakes: Isolation Forest F1 0.34, LSTM autoencoder 0.40 vs Z-score 0.28 · Tyres: RNN and LSTM best on lap times; pit window above chance